BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 238 · Thinking abstractly: models and reality

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is unit testing?

Q2. What is a precondition?

Q3. What do vertices and edges represent in a graph?

Q4. Why is an unweighted route graph not enough to identify the fastest journey?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

abstraction — Retain relevant characteristics, ignore irrelevant ones.

model — Simplified representation serving a purpose.

attribute — Property stored about an entity.

assumption — Condition accepted in constructing the model.

limitation — A question the simplified model cannot answer.

New learning

Choose a purpose — Decide whether a particular book may be loaned.

Keep essential facts — ISBN identifies the book, and Available indicates whether it can be borrowed.

Evaluate omissions — The model cannot answer questions about book weight or appearance because those attributes are omitted.

Worked example

Model a bus network for finding the route with the fewest stops.

1. Predict first — Model each bus stop as a vertex and each permitted connection as an edge.

2. Model and check — Vehicle colour is irrelevant. To find fastest rather than fewest-stop routes, add journey-time weights.

We Do · Practise with guidance

Model the average of a class's valid test marks.

```
average = sum(validMarks) / count(validMarks)
```

Predict first

Only valid marks and the number of entries are needed to compute the mean.

Model and check

Define how to handle missing results. Other pupil attributes need not be stored for this calculation.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Identify two relevant and two irrelevant attributes for a stock checker.

Question 2

Give a simple abstraction for detecting school room timetable clashes: three needed fields and two omitted details.

Question 3

Why should a model not contain every real-world detail? Give an example where a previously omitted detail becomes relevant.

Exit Ticket · Answer before checking

A good abstraction must record every real-world property.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Testing a small module or function independently.

Q2. An input requirement that must hold before execution.

Q3. Objects/locations and connections.

Q4. It lacks travel-time weights.

You Do pages · Independent questions

Q1. [4] Relevant: product identifier and quantity [2]; irrelevant to stock: label colour and shop music [2].

Q2. [6] Room identifier [1], lesson start [1], lesson end [1]; compare overlapping times in same room [1]; omit poster colour [1] and desk appearance [1].

Q3. [6] Reduces storage [1], complexity [1] and irrelevant data exposure [1]; keeps purpose clear [1]; journey duration becomes relevant when switching from fewest stops to fastest route [2].

Exit Ticket · Model answer

False. It records properties relevant to the model's purpose.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding